

CLEAN-REPO PRE-GO GATE — the definitive checklist before ANY external (flagship Forcing+Evidence)

Keeper (hub + final audit), 2026-08-09. The flagship PASSED Cal's hostile read (K1313). Casey approved both pending decisions (the two flagship wordings + Grace's PMNS dedup). This is the living gate: every box must be TRUE before external. Nothing external until all checked + Casey's explicit GO. This is the standing "clean the repo" gate (task #104).

Status legend: [] open · [~] in progress · [x] done+verified

A. Paper fixes (Cal's 4, from the hostile read) — Lyra applies → Cal re-confirms

- ☒ A1. $\cos \psi = 5/\sqrt{34} \rightarrow V_{cb}$ marked CANDIDATE, both gaps named — DONE in flagship v0.2.3 (Lyra: value gap 6–10° vs 2.4° + innocence gap named). (Owner: Lyra)
- ☒ A2. "forced modulo one datum = GR-level" qualified — DONE v0.2.3 (GR-level in input economy, carries the I.5 residuals). (Owner: Lyra)
- ☒ A3. parity forces the SHAPE ($\sin^2 \theta_W = 3/13$ PD) + CP existence not value — DONE v0.2.3. (Owner: Lyra)
- ☐ A4. Cal re-confirms A1–A3 applied in the file. (Owner: Cal — pending; fixes are in v0.2.3, Cal re-reads.)

B. Clean-repo artifacts — Casey APPROVED (2026-08-09), execute

- ☐ B1. The two flagship wordings (Casey approved): data/bst_this_is.md (α = charge-count 137 + curvature correction, "Wyler integral" attribution retired) + README Wyler headline/homage (honor his SO(5,2) *insight* + the direction he reached toward; his exact $\text{Vol}^{(1/4)}$ formula retired; α presented as $N_{\max}$ charge-count + curvature, not the Wyler formula). (Owner: Grace/Lyra to stamp per Grace's already-drafted wording; Keeper verifies bodies match banners — no half-done sweep.)
- ☐ B2. PMNS dedup (Casey approved): keep ONE clean geometric form per angle ($\theta_{23} \rightarrow 4/7$, $\theta_{13} \rightarrow 1/45$, $\theta_{12} \rightarrow$ the picked one of {4/13, 42/137}); mark the compound (44/45-suffixed) forms superseded, never silent-delete; fill/remove empty predicted-value cells. Ties task #84 (pointer-integrity). (Owner: Grace)
- ☐ B3. Millennium PDFs verified current — NOT fully closed (Cal re-confirm, 2026-08-09). The 17 numbered Millennium PDFs are current ☒. BUT Cal's broader scan caught stragglers the subset-report missed: FIVE Riemann-family papers have source newer than their PDF (if any carries a Proof→Attempt edit, the compiled PDF may not show it), and BST_RH_Weil_Positivity_Proof has NO PDF at all (the paper with the LaTeX error — build failed or never ran). FIX: rebuild + re-read those five, confirm RH_Weil actually compiles, OR mark each explicitly "internal/superseded, out of scope" — each verified by opening the compiled artifact. "Content-ready isn't cleared." (Owner: RH-paper owner / Elie build + Cal re-read.)
- ☐ ★ B4. RH "PROVED"-title hazard (Cal, HIGH — hard blocker, K1316). BST_Why_Geometry_Proves_Riemann (title says "...Proves Riemann") and Koons_Riemann_BST_2026 ("the conjecture has been PROVED via three independent routes") — a serious reader hitting either discredits the whole repo on sight. Reframe both to "Attempt" (K940, task #103) + rebuild, OR mark superseded + remove PDFs; each verified by re-opening the compiled artifact. Confirm the two "Superseded by Paper #103" Riemann files are scoped out. Millennium outline (reframed but stale) → rebuild or scope out. (Owner: pandoc/honesty lane + Cal re-verify.) Non-negotiable before external.

C. Consistency sweep (Keeper final audit) — no banner without a matching body

- ☐ C1. The V_{us} split + F85-radial-Derived + mixing-Identified tiers are consistent across the flagship, data/bst_constants.json, and the registry (no stale "all CKM derived" / " $V_{cb} = 4/79$ " anywhere). V_{cb} consistency web (K1316): V_{cb} Candidate ☒ A_Wolf → Candidate ($A = |V_{cb}|/\lambda^2$, currently mis-tagged Derived in 2 entries); k1001/k1003 " V_{cb} Derived" prose corrected-forward (Grace, done). (Owner: Keeper + Grace)

- ★★ C1-CRITICAL — the CP gate was closed on the CKM side but NOT the PMNS side (Keeper C1 sweep, 2026-08-09). The binding CP gate (existence-only, NO J, NO δ VALUE) applies to *both* sectors, but the lepton side still leaks:
 - * Registry T2536 — “ $\delta_{\text{PMNS}} \cos^2 \delta = 45/49 + J_{\text{PMNS}} \approx 0.0338$ — DERIVED (forward)”: a leptonic δ -value AND a J-value, tagged Derived. Same binding-gate violation Grace just purged on the CKM side.
 - * data/bst_predictions.json — “ δ_{PMNS} (leptonic CP phase) — D”.
 - * FIX (parallel to Grace’s CKM purge): the $\delta_{\text{PMNS}} / J_{\text{PMNS}}$ values $\rightarrow$ existence-only, off (no number, no agreement-%); keep only “leptonic CP exists (forced)”. If the $\cos^2 \delta = 45/49$ sum-rule is a real *structural* relation, keep it Structural/candidate WITHOUT the displayed value. (Owner: Grace [data] + registry owner; Keeper re-verifies.)
- C1-MODERATE — PMNS θ_{12}/θ_{13} mis-tagged “D” in live data. data/bst_constants.json tags “PMNS $\sin^2 \theta_{12}$ (solar)” and “PMNS $\sin^2 \theta_{13}$ (reactor)” as D. Per K1313/K1314 the PMNS *structure* is Derived but the *values* stay Identified/gated on the kernel (Grace’s own blind score). Re-tier both D $\rightarrow$ I. ($\theta_{12} = 3/10$ is the geometric-primary FORM, Identified value.) (Owner: Grace.)
- C1-MODERATE — registry “Proved” on CKM/PMNS mixing theorems (T919, T1254, T1259, T1474, T2510, T2535 all say “Proved”). The registry Proved column is not the rigorous D/PD/I/C/S tier (task #91), but an external reader reads “Proved” as Proved. Annotate the CKM/PMNS mixing rows with the honest current tier (Candidate/Identified/Structural), or add the “Proved-column $\neq$ rigorous-tier” header note. (Owner: registry governance, ties #91.)
- ☐ C2. CP: existence-only, NO J value, no δ value, $\sim O(100\times)$ noted — everywhere (paper, data, registry). (Owner: Keeper)
- ☐ C3. #79 stated as “forced modulo one datum” with Leg B(ii) at its K1312 tier (forced-at-structure + named real $\rightarrow$ complex residual), NOT over-claimed as Derived-end-to-end. (Owner: Keeper)
- ☐ C4. No “600+ predictions” volume-lead; α “not Wyler”; $\sin^2 \theta_W$ PD; dead gravity-tie absent — re-confirmed post-edits (Cal’s gates stay green after the wording changes). (Owner: Keeper + Cal)

★ C-SWEEP FINDINGS (Keeper, 2026-08-09) — real leaks in EXTERNAL-facing files (the flagship is clean; the repo front-door is NOT)

The paper passed Cal’s gate, but the repo (README + data/) still carries retired/over-claimed forms a hostile external reader would hit first. Since the public stance is “the math’s on GitHub,” these are GO-blocking. Routed, not fixed (ownership: Grace=data, README=Grace/Lyra + Casey tone call). - [x] ★ CRITICAL — CP J-VALUE leak — CLOSED 2026-08-09 (Grace, ratified K1314). Both J_CKM ledger entries neutralized + 3 README rows purged ($\gamma = \arctan \sqrt{5}$, $A^2 \lambda^6 \bar{\eta}$ @0.3%, summary CP row). Repo now reads CP existence-only, magnitude/phase off, no values. ALSO ruled K1314: $\bar{\eta}$ value $\rightarrow$ OFF (CP-violating), $\bar{\rho} \rightarrow$ Identified (Grace to apply). Original finding retained below for record: - README.md:152 — table row “J_CKM (Jarlskog) | $A^2 \lambda^6 \bar{\eta}$ | 3.07×10^{-5} | 3.08×10^{-5} | 0.3%” — a *retired reverse-fit* (Wolfenstein $A^2 \lambda^6 \bar{\eta}$, demoted D $\rightarrow$ I per K684; data file itself calls it “not a forward derivation”) shown as a 0.3% prediction on the front page. - README.md:556 — “CKM CP phase | $\gamma = \arctan(\sqrt{5})$; $J = \sqrt{2}/50000$ | 0.6%” — the retired $\gamma = \arctan(\sqrt{5})$ reverse-fit (K683) AND a J value, both stale. - data/bst_constants.json J entries (const ~ 2866 , ~ 5384) display $J = 3.072e-5$ “(observed $3.08e-5$)” and $J = g^4 c_2^2 / (n_C \cdot N_{\text{max}}^5)$ — tier-labeled I with honest mechanism, but the *value + observed-comparison still display*. - FIX: purge all J/ δ VALUES + agreement-% from README + data external displays; state CP existence-only, magnitude $\sim O(100\times)$ off (per K1304 hard rule). Retire the $\gamma = \arctan(\sqrt{5})$ row. (Owner: Grace + README owner; Keeper re-verifies.) This is the same binding gate the whole paper rests on — the repo must not contradict it. - [] MODERATE — $\cos \psi$ marked “Derived” in live data. data/bst_26_tier_map.json:115 says “Direction $\cos \psi = 5/\sqrt{34}$ (Derived, F384)” — contradicts K1313 ($\cos \psi \rightarrow V_{cb}$ is Candidate; projection open). Downgrade to Candidate; keep V_{cb} Structural/down-only. (Owner: Grace.) - [] FLAG (Casey tone call) — “600+ predictions” volume-lead. data/bst_this_is.md:19, README.md:84, README.md:578 lead with “600+ predictions.” The flagship deliberately avoids this (Cal gate). At minimum qualify to “600+ predictions, honestly tiered (most Identified/Structural, few Derived)” so the volume-lead doesn’t read as 600 Derived results. (Owner: Casey’s call on README tone; Keeper recommends the qualifier.) - Note: data/bst_this_is.md:9 α line is already honest (charge-count 137 + curvature n_C/N_{max} , “Wyler’s exact formula didn’t hold, direction right”) — B1 partially

in there; verify README:127/428/658 match.

D. Final GO

- ☐ D1. All of A–C checked. Cal green. Keeper final PASS.
- ☐ D2. Casey’s explicit GO. (Standing rule: nothing external — Zenodo/arXiv/outreach — without Casey’s word. Never push without approval.)

Notes

- The paper does NOT wait on the K1012 kernel — the mixing values ship honestly Identified/gated; the kernel is the later Identified→Derived upgrade.
- This gate covers the flagship (Forcing+Evidence). The Finster/CFS bridge note (task #102) is a SEPARATE, later, still-gated item — not part of this GO.
- Keep this file updated as boxes close; it is the single source of truth for “are we clear to ship.”

— Keeper, clean-repo pre-GO gate, 2026-08-09. Flagship passed hostile read; Casey approved the 2 wordings + the PMNS dedup. Gate = A (Lyra’s 4 fixes + Cal re-confirm) + B (wordings + dedup + PDFs) + C (Keeper consistency sweep) + D (Cal green, Keeper PASS, Casey GO). Nothing external until all checked. Nothing pushed.